Hospital Analytics Dashboard

End-to-End SQL + Power BI Project

A Data Analytics Case Study

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1. Project Overview

This project focuses on building an end-to-end hospital analytics dashboard using SQL and Power BI. The objective is to transform raw healthcare data into meaningful insights that support decision-making across operations, finance, and patient management.

The dashboard provides a comprehensive view of hospital performance, including revenue analysis, doctor efficiency, patient behavior, and financial tracking. It demonstrates how data can be used to identify inefficiencies and improve overall healthcare service delivery.

2. Why This Project?

Healthcare institutions generate large volumes of data daily but often lack structured reporting and insights.

This project addresses key real-world challenges such as:

- Lack of visibility into revenue and payment collections
- Uneven performance among doctors
- Limited understanding of patient trends
- Difficulty in tracking operational efficiency

By solving these problems, the dashboard helps simulate real-world decision-making scenarios in hospital management.

3. Industry Use Cases Covered

This project replicates several real-world healthcare analytics use cases:

- **Revenue Analysis** → Tracking total revenue, pending, and failed payments
- **Doctor Performance** → Identifying top and low-performing doctors
- **Patient Insights** → Understanding demographics and treatment patterns
- **Operational Efficiency** → Monitoring appointments and workload trends
- **Financial Monitoring** → Evaluating payment success rate and collection gaps

4. Tools & Technologies Used

SQL

- Data cleaning and transformation
- Handling missing values and inconsistencies
- Creating structured tables for analysis

Power BI

- Data modeling (Star Schema)
- DAX measures for KPIs and calculations
- Interactive dashboard creation

5. Why SQL + Power BI (Instead of Only Power BI)?

Using SQL and Power BI together reflects a real-world data workflow:

- SQL is used for **data preparation** (cleaning, transformation, structuring)
- Power BI is used for **analysis and visualization**

This separation improves:

- Performance
- Scalability
- Clarity in data modeling

It also demonstrates the ability to work with both **backend (SQL)** and **frontend (BI tools)**.

6. Project Goals

The key objectives of this project were:

- Build a structured **star schema data model**
- Create meaningful KPIs for hospital performance
- Design **interactive dashboards** for analysis
- Identify **revenue and operational gaps**
- Present insights in a **clear and business-friendly way**

7. Learning Outcomes

Through this project, the following skills were developed:

- Data cleaning and preprocessing using SQL
- Data modeling (fact and dimension tables)
- Writing DAX measures for business metrics
- Designing user-friendly dashboards
- Converting raw data into actionable insights

8. Dashboard Breakdown

Page 1: Hospital Overview

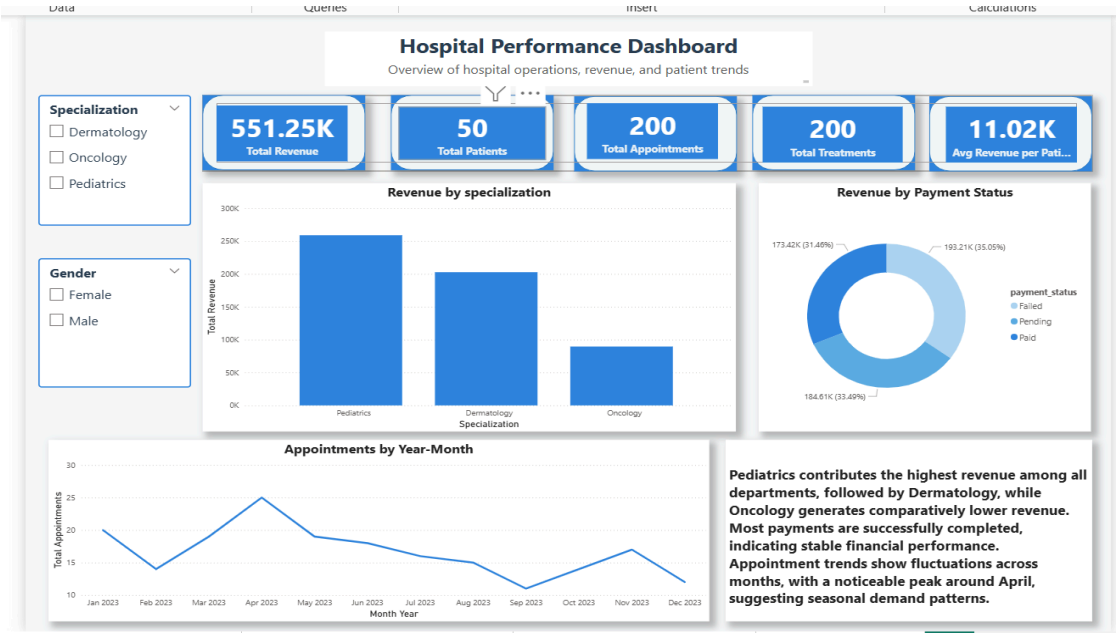
This page provides a high-level summary of overall hospital performance.

Key Features:

- KPIs: Total Revenue, Patients, Appointments, Treatments
- Revenue by specialization
- Payment status distribution
- Monthly appointment trends

Key Insight:

A significant portion of payments remains pending or failed, indicating opportunities to improve collection efficiency.



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Page 2: Doctor Performance

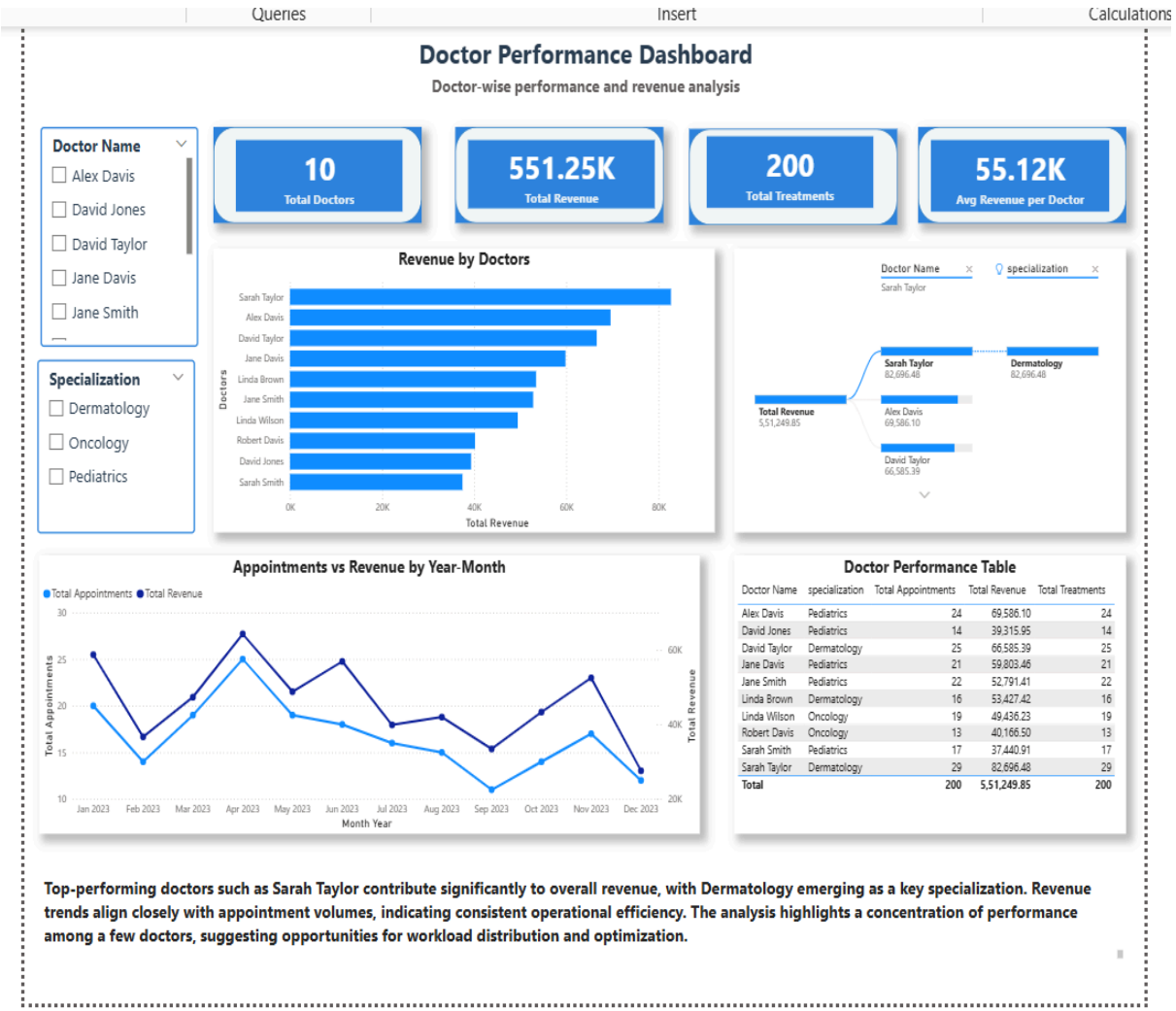
This page focuses on evaluating doctor-level performance.

Key Features:

- Revenue by doctor (Top performers)
- Decomposition tree for deep analysis
- Appointments vs Revenue trend
- Detailed performance table

Key Insight:

A small group of doctors contributes a large portion of revenue, indicating performance concentration and potential workload imbalance.



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Page 3: Patient & Treatment Insights

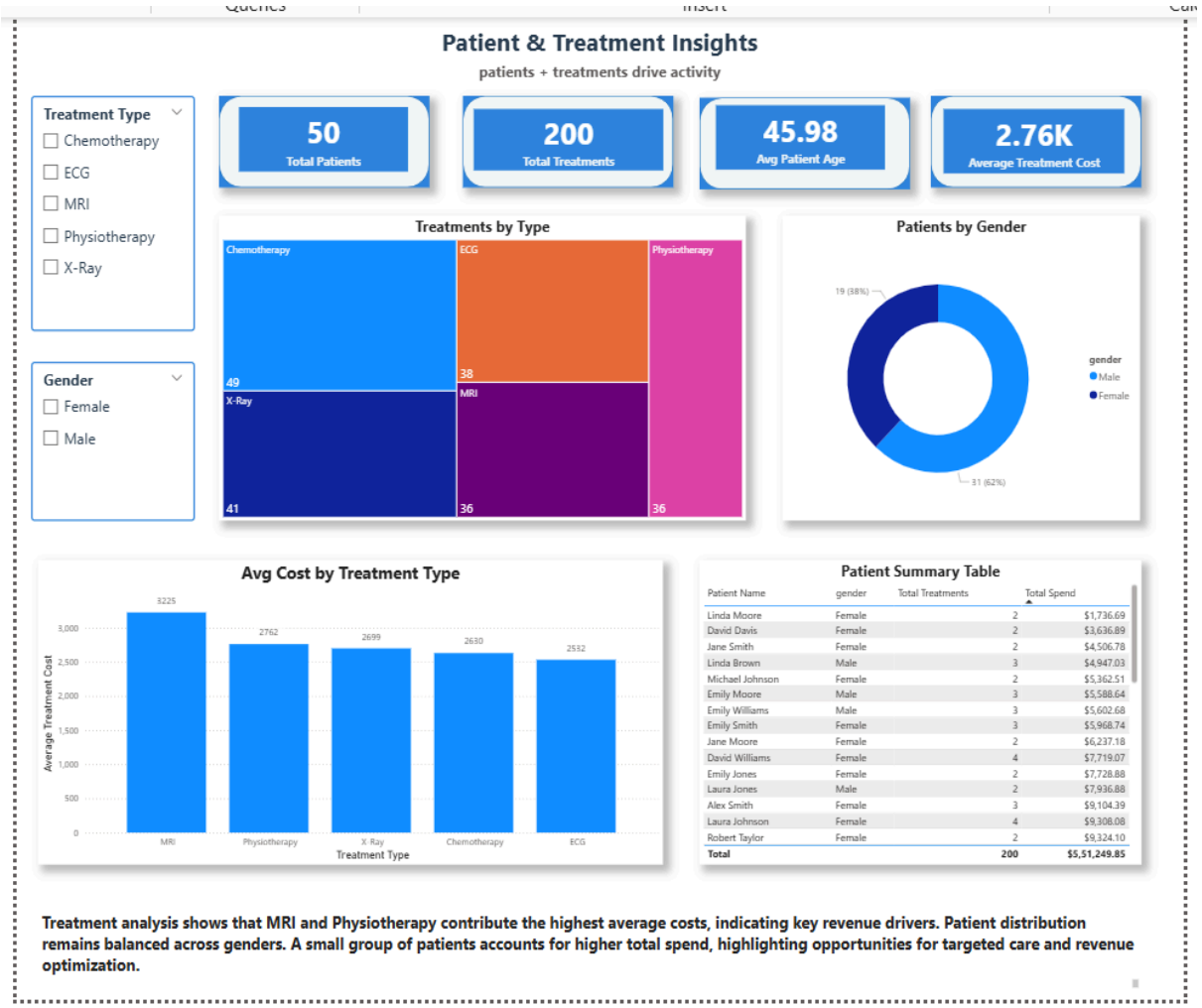
This page analyzes patient demographics and treatment patterns.

Key Features:

- Average patient age
- Average treatment cost
- Treatment distribution (treemap)
- Patient spending analysis

Key Insight:

Certain treatments contribute significantly to overall cost and revenue, highlighting key service areas.



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Page 4: Financial Analysis

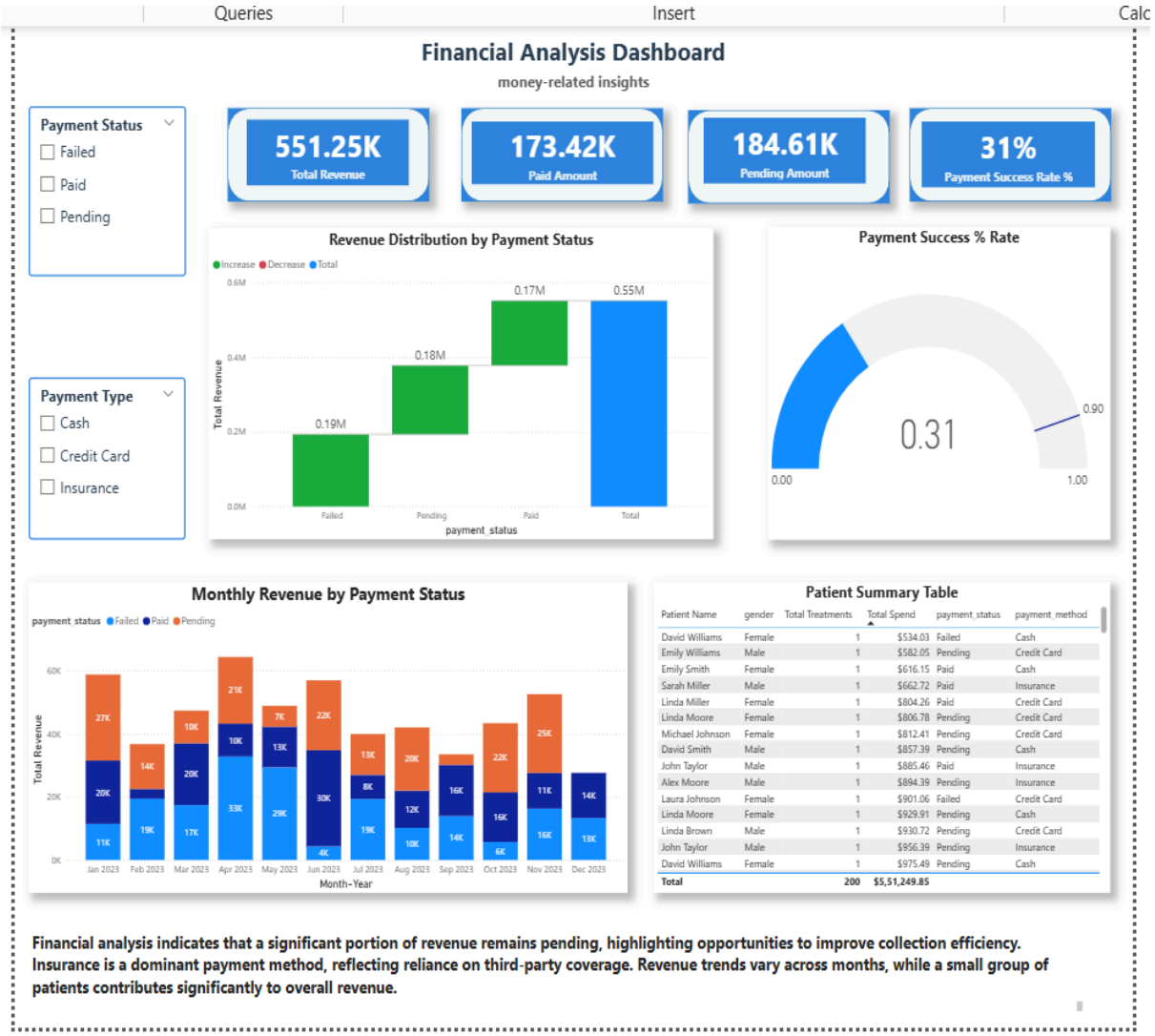
This page focuses on revenue flow and payment behavior.

Key Features:

- Total, Paid, and Pending revenue
- Payment success rate
- Revenue breakdown (waterfall chart)
- Monthly revenue trends
- Payment method analysis

Key Insight:

A considerable portion of revenue remains uncollected, indicating the need for improved billing and payment processes.



9. Conclusion

This project demonstrates the development of a comprehensive end-to-end data analytics solution, starting from raw, unstructured healthcare data and transforming it into meaningful, actionable insights. By leveraging SQL for data cleaning and transformation, and Power BI for data modeling and visualization, the project reflects a real-world analytics workflow used in industry. The implementation of a structured star schema, along with carefully designed DAX measures, ensures efficient data handling and accurate reporting across multiple dimensions.

Beyond technical execution, the project emphasizes strong business understanding by addressing key operational and financial challenges faced by healthcare institutions. The dashboard provides clear visibility into hospital performance, enabling stakeholders to monitor revenue trends, evaluate doctor efficiency, understand patient behavior, and assess financial health. Through interactive visuals and well-defined KPIs, the solution supports data-driven decision-making and highlights areas that require attention.

A critical insight derived from the analysis is the presence of a significant proportion of pending and failed payments, indicating opportunities to improve billing processes and collection efficiency. Additionally, the concentration of revenue among a limited number of doctors suggests the need for better workload distribution and performance optimization. Treatment-level analysis further reveals that certain services act as primary revenue drivers, offering opportunities for strategic focus and resource allocation.

Overall, this project not only showcases technical proficiency in SQL and Power BI but also demonstrates the ability to translate data into meaningful business insights. It highlights the importance of combining analytical thinking with domain understanding to solve real-world problems. The dashboard serves as a scalable foundation that can be further enhanced with predictive analytics, real-time data integration, and advanced performance metrics, making it a practical and industry-relevant solution in the field of healthcare analytics.